Type MS · Identify skin structures and state what each does

Answer 1 of 8 · Question 1 part b.iii · 2 marks

2024 JUN · Paper 01 · find part b.iii in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · Identify skin structures and state what each does

Answer 2 of 8 · Question 5 part a · 3 marks

2020 JAN · Paper 02 · find part a in this mark scheme extract

Number				
5	(a)	• (layer 1) epidermis(1) • (layer 2) dermis (1) • (layer 3) fat/subcutaneous fat/fat layer/fat tissue/adipose layer/adipose tissue		3
	(b)	**The only correct answer is A** blood vessels and sebaceous glands ***B** is incorrect because these are in the third layer* ***C** is incorrect because these are in the third layer* ***D** is incorrect because the fat tissue is the third layer*		1
	(c)	• skin/epidermis/tissues/blood vessels damaged/open to air(1) • bacteria can enter bloodstream/body(1)		2
	(d)	Any four from: • platelets accumulate at wound/damaged area(1) • enzyme/thromokinase/thromboplastin released (1) • prothrombin converted to thrombin(1) • fibrinogen to fibrin(1) • mesh traps red blood cells(1) • scab/clot formed(1) • prevents entry of bacteria / microorganisms(1)		4

Total for Question 5 = 10 marks

Type MS · Identify skin structures and state what each does

Answer 3 of 8 · Question 5 part b · 1 mark

2020 JAN · Paper 02 · find part b in this mark scheme extract

Number				
5	(a)	• (layer 1) epidermis(1) • (layer 2) dermis (1) • (layer 3) fat/subcutaneous fat/fat layer/fat tissue/adipose layer/adipose tissue		3
	(b)	**The only correct answer is A** blood vessels and sebaceous glands ***B** is incorrect because these are in the third layer* ***C** is incorrect because these are in the third layer* ***D** is incorrect because the fat tissue is the third layer*		1
	(c)	• skin/epidermis/tissues/blood vessels damaged/open to air(1) • bacteria can enter bloodstream/body(1)		2
	(d)	Any four from: • platelets accumulate at wound/damaged area(1) • enzyme/thromokinase/thromboplastin released (1) • prothrombin converted to thrombin(1) • fibrinogen to fibrin(1) • mesh traps red blood cells(1) • scab/clot formed(1) • prevents entry of bacteria / microorganisms(1)		4

Total for Question 5 = 10 marks

Type MS · Identify skin structures and state what each does

Answer 4 of 8 · Question 5 part c · 2 marks

2020 JAN · Paper 02 · find part c in this mark scheme extract

Number				
5	(a)	• (layer 1) epidermis(1) • (layer 2) dermis (1) • (layer 3) fat/subcutaneous fat/fat layer/fat tissue/adipose layer/adipose tissue		3
	(b)	**The only correct answer is A** blood vessels and sebaceous glands ***B** is incorrect because these are in the third layer* ***C** is incorrect because these are in the third layer* ***D** is incorrect because the fat tissue is the third layer*		1
	(c)	• skin/epidermis/tissues/blood vessels damaged/open to air(1) • bacteria can enter bloodstream/body(1)		2
	(d)	Any four from: • platelets accumulate at wound/damaged area(1) • enzyme/thromokinase/thromboplastin released (1) • prothrombin converted to thrombin(1) • fibrinogen to fibrin(1) • mesh traps red blood cells(1) • scab/clot formed(1) • prevents entry of bacteria / microorganisms(1)		4

Total for Question 5 = 10 marks

Type MS · Identify skin structures and state what each does

Answer 5 of 8 · Question 8 part a.i · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
8	(a) (i)	absorbs uv light; reduces/prevents risk of skin cancer;	Allow protects against (UV) radiation	2
	(ii)	eye/ retina/ choroid/ red blood cell/ haemoglobin ;		1
	(b) (i)	mother x father Hh hh; gametes H h h; fertilisation Hh hh; phenotype white forelock normal;	Reject diagrams that show sex linkage	4
	(ii)	chance of producing a boy is 0.5/½; chance of producing offspring with condition 0.5/½; chance of producing boy with condition is ½ × ½ = ¼;	Allow 50% or equivalent anywhere in calculation Full marks for correct final answer	3
	(c)	Any 4 from different (genetic) code produced; different order of amino acids; codes for different proteins/enzymes; enzyme substrate complex not formed; causes change in pigment;	Allow change in base sequence/genetic code Allow any reference to change in amino acids	4

Total 14 marks

Type MS · Identify skin structures and state what each does

Answer 5 of 8 · Question 8 part a.i · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Identify skin structures and state what each does

Answer 6 of 8 · Question 3 part a · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Number				
3	(a)	any two from • protection/named example/barrier against pathogens; • sensory reception/named example; • excretion		2
	(b)	• looks red; • vasodilation/more blood in skin <u>capillaries</u>/blood flows closer to skins surface; • allows more heat loss/heat energy released; • to lower body temperature;		4
	(c) (i)	• as temperature increases, rate of sweating increases; • up until 34°C/ between 32 °C - 35.9 °C, little/no change; • increases, rapidly/significantly from 35.9 °C - 38 °C;	Allow any temperature between 34°C and 35.9 °C for 35.9 °C e.g. between 32°C and 34 °C	2
	(ii)	• absorb sweat with cotton wool/absorbent paper; • fixed area of skin/under arm pits; • reference to measuring mass of absorbent material before and after; or • measure time; • for cobalt chloride paper to change colour;		2
	(d)	• enzymes affected/to maintain enzyme activity/ref to body temperature being optimum for enzymes; • too high a temperature denatured/prevent denaturation; • too low, inactive/less active/prevent enzymes becoming inactive;		3

Total 13 marks

Type MS · Identify skin structures and state what each does

Answer 6 of 8 · Question 3 part a · 2 marks

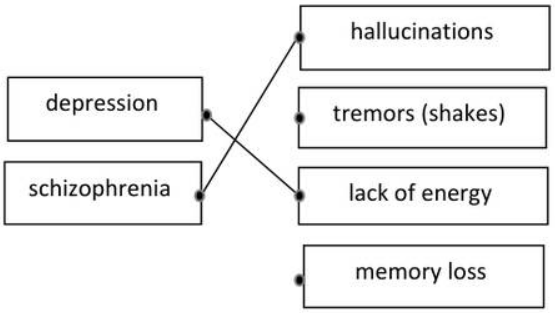
2022 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify skin structures and state what each does

Answer 7 of 8 · Question 2 part d · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · Identify skin structures and state what each does

Answer 7 of 8 · Question 2 part d · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify skin structures and state what each does

Answer 8 of 8 · Question 1 part b.i · 2 marks

2024 JUN · Paper 01 · find part b.i in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks